

Applied AI Engineer specializing in LLM systems, retrieval-augmented generation, and multi-agent pipeline design. Experienced in fine-tuning transformer models, constructing synthetic training corpora, and deploying AI systems that operate reliably in production environments. Designs end-to-end AI architectures that combine deterministic business logic with LLM reasoning — including multi-stage retrieval pipelines, schema-aware orchestration layers, and structured validation for regulated domains. Brings applied research depth alongside hands-on engineering across financial workflows and enterprise systems, demonstrated through production RAG deployments, and full-cycle model development from synthetic data generation to inference.

EDUCATION

Master of Science in Data Science, University of Colorado Boulder	Fall 2025 GPA: 3.6
Bachelor of Science in Business Marketing, Metropolitan State University of Denver	May 2021 GPA: 3.4

SKILLS

LLM Engineering	Prompt engineering, instruction tuning, fine-tuning (LoRA/QLoRA), synthetic data generation, transformer architectures (encoder-decoder, attention), multi-agent orchestration, inference optimization
NLP & Retrieval	BERT, sentence transformers, semantic search, RAG pipeline design, re-ranking, chunking strategies, embedding pipelines
Machine Learning	Neural networks, tree-based models, time-series forecasting, feature engineering, model evaluation, interpretability
AI Data Architecture	Vector databases (Pinecone, Weaviate, FAISS, ChromaDB), data schema design, ETL for training corpora, synthetic dataset curation, hybrid retrieval over structured and unstructured data sources
Engineering & MLOps	Python, SQL, Hugging Face, PyTorch, FastAPI, Docker, Google Cloud Platform (GCP)

RESEARCH, PROJECTS & EXPERIENCE

Lantern Intelligence: AI Accounting Assistant (SaaS)	Aug 2025 – Present
AI Systems Engineer	

- Built a RAG-based AI financial advisory platform using FastAPI, SQLite, enabling natural-language interaction with structured financial data across accounts payable, receivable, payroll, and forecasting.
- Architected a modular retrieval pipeline combining deterministic SQL execution with semantic search over benchmark and definitional knowledge, ensuring grounded, auditable outputs for regulated financial workflows.
- Designed a multi-database architecture representing distinct company financial profiles, enabling the same query to produce contextually differentiated responses across client scenarios.

Abductive Event Reasoning — SemEval 2026 Task 12	Aug 2025 – Dec 2025
CU Boulder Research Group	

- Built an abductive reasoning system to infer plausible missing events from multi-document evidence using multi-stage retrieval, candidate ranking, and hypothesis-scoring pipelines.
- Designed baseline evaluation workflows for reproducible, systematic comparison of abductive inference model variants.
- Translated theoretical reasoning frameworks (abductive NLI, narrative cloze) into benchmarkable AI systems within a collaborative research group.

Applied ML Research	2024 – 2025
Independent and Academic Research	

- Interview Outcome Prediction:** Built multimodal ML models combining acoustic speech features and transcribed text to predict interview performance, with emphasis on interpretable and explainable modeling.
- Housing Price & Rent Forecasting:** Developed non-linear time-series models using macroeconomic indicators; performed feature engineering and diagnostic analysis to address autocorrelation and non-stationarity.

Finance & Management	2014 – 2018
Multiple GM Dealerships, Denver CO	

- Structured complex financial deals involving credit evaluation and lender coordination under regulatory constraints — developing domain fluency in financial products and risk assessment that directly informs current AI finance work.